Nandana Dileep

Bangalore, India (Open to Relocation) | nandanadileep29@gmail.com | +91 8301939350
nandanadileep.com | nandana-dileep | nandanadileep

PROFILE

AI Systems Engineer focused on making agentic software reliable at the boundary between model decisions and tool execution. At JLR Technology & Business Services India, building an AI-native spec-driven engineering platform adopted by a major vehicle program. Creator of Mycelium, an open-source agent runtime with 20.0k+PyPI downloads, and an MLX-based LLM serving engine.

PROFESSIONAL EXPERIENCE

AI Engineer, *JLR Technology & Business Services India, SDV*, Bangalore, India — July 2024 - Present

- **Sarvam (Spec-Driven Development):** Core developer on an 8-person SDV team building an AI-native Everything-as-Code platform for automotive requirements, HLD/LLD, tests, and compliance. Developed version-controlled AsciiDoc workflows, GitLab CI/CD quality gates, AWS Bedrock-assisted authoring and review, and a knowledge-graph digital thread; adopted by a major vehicle program.
- **Production ML Optimization:** Deployed SMOTE-balanced ensemble classifier identifying underutilized test rigs, achieving **18% cost reduction** through asset consolidation (processed 500K+ telemetry events daily).
- **GNSS Localization:** Engineered a C++ GNSS processing pipeline using the GTSAM factor graph library, significantly improving vehicle coordinate estimation accuracy. Presented original research at JLR Tech Symposium.

Software Development Engineer Intern, *Amazon*, Bangalore, India — May 2023 - July 2023

- **Distributed Systems:** Designed and deployed a Latency Analysis Service for Alexa using AWS Lambda, Coral, and OpenSearch; completed production deployment ahead of schedule.
- **Test Automation:** Built end-to-end testing framework using JUnit 5 and Mockito, increasing code reliability and reducing deployment rollback rates.

RESEARCH & PROJECTS

Meridian: AI Hiring Automation, *JLR Global Techathon* — Winner, 2026

- Built a multi-tenant platform automating hiring from JD ingestion and candidate skill-gap analysis through recruiter review and adaptive technical assessments; selected as the winner among teams from the UK, China, and Hungary.
- Converted job descriptions into skill-ontology graphs and overlaid resume evidence as present/partial/gap nodes in an interactive D3 recruiter view; engineered 5×5 topic-tree routing, 1PL Rasch ability estimation, bank-first question serving, and embedding/lexical deduplication with human-gated, auditable decisions.

Mycelium: Reliable Tool Execution for AI Agents, *v1.31.0 · 20.0k+downloads* — Docs, GitHub, PyPI

- Built a framework-independent Python runtime that wraps consequential agent tool calls in durable transition envelopes; retries, crashes, and concurrent redispatches return stored results, poll in-flight work, reconcile provider state, or fail closed instead of blindly repeating mutations.
- Published the Python SDK on PyPI (v1.31.0, **20.0k+** downloads) with YAML auto-instrumentation, LangGraph and CrewAI integrations, Redis/PostgreSQL ledgers, and 800+ tests covering process crashes, multi-worker races, and storage outages.

KAIROS / Auto-Agent: Autonomous Background Coding Agent — GitHub

- Built an autonomous daemon that gathers context from git history, TODOs, modified files, stale pull requests, and environment configuration, then uses an LLM scheduler to decide whether to act or sleep.
- Added presence-aware notifications and nightly "AutoDream" memory consolidation, compressing observations into persistent project memory while pruning contradictions.

vLLM-Inspired LLM Serving Engine on Apple Silicon — GitHub

- Built an MLX-based serving engine from scratch with PagedAttention-inspired KV cache management, continuous batching, chunked prefill, prefix caching, and asynchronous request scheduling.
- Exposed OpenAI-compatible chat completions and benchmarked the implementation against vLLM and llama.cpp; published an architectural deep dive.

Identiti: Personal AI Memory Layer with Knowledge Graphs — Live, GitHub

- Built a production full-stack memory system that extracts confidence-scored identity entities from conversations and stores their relationships in a Neo4j knowledge graph.

- Implemented React/Vite, Flask, Redis, Supabase authentication, per-user field-level encryption, and portable "Memory Card" export; deployed on Vercel and Render.

WRITING

"Checklist to Prevent All the Agent Failure Modes" — Medium, 2026

Systematic framework for diagnosing and preventing failure modes in agentic LLM systems.

"Building a vLLM-Inspired LLM Serving Engine on Apple Silicon with MLX" — Medium, 2026

Architectural deep dive into production LLM serving: PagedAttention-inspired KV cache management, continuous batching, and async scheduling built from scratch on Apple Silicon.

"Swapped Every Part of nanoGPT With Modern Alternatives" — Medium, 2026

Component-by-component ablation study of modern transformer techniques on nanoGPT; benchmarked each substitution independently and published results.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL

Agentic Systems: Tool Calling, MCP, Agent Memory, Runtime Guardrails, RAG, Knowledge Graphs

ML/LLMs: PyTorch, Transformers, PEFT, MLX, vLLM, FAISS, Scikit-learn

Backend/Infrastructure: FastAPI, Flask, React, Redis, PostgreSQL, Neo4j, Docker, AWS, GitHub Actions, GitLab CI

EDUCATION

Indian Institute of Technology Madras (IIT Madras) - Bachelor of Technology — Aug 2020 - May 2024

Major: Civil Engineering

PUBLICATIONS

Precise Position Estimation using Factor Graph Optimization

Jaguar Land Rover Tech Symposium, June 2024. Original research on GNSS localization using factor graph methods in C++ with GTSAM.